Video Game Sales Analysis

Mini Project – Data Analysis and Visualization

Team Members

- Aditya Gupta (251090051560)
- Shubham Agarwal (251090051934)

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1. Introduction

The video game industry is one of the fastest-growing entertainment sectors globally, generating billions of dollars in revenue annually. This project explores a comprehensive dataset of video game titles, their critic scores, release years, platforms, and regional sales figures to uncover trends, patterns, and insights within the industry.

The analysis aims to answer key business and analytical questions such as:

- Which game genres generate the highest total sales worldwide?
- Which gaming consoles have dominated the market over the decades?
- How are sales distributed across regions (North America, Japan, Europe, Others)?
- Is there a meaningful relationship between critic scores and commercial success?
- How have video game sales evolved year over year?
- Which publishers and developers lead in terms of total global sales?

2. Dataset Description

2.1 Source

The dataset consists of two CSV files provided for this lab case study:

- **Video_Games.csv** – Contains metadata about video game titles including title, console, genre, publisher, developer, critic score, and release date.
- **Sales.csv** – Contains sales figures (in millions of units) split by region: North America (na_sales), Japan (jp_sales), PAL/Europe (pal_sales), and other regions (other_sales), along with a total_sales column.
- Both datasets share a common key column Unique_ID, which was used to merge them for the analysis.

2.2 Dataset Overview

Attribute	Video_Games.csv	Sales.csv
Total Records	64,016 rows	64,016 rows
Columns	9 columns	6 columns

2.3 Key Variables

Variable	Description	Type
title	Name of the video game	String
console	Gaming platform (PS2, Xbox, PC, etc.)	String
genre	Game category (Action, Sports, RPG, etc.)	String
publisher	Company that published the game	String
developer	Company that developed the game	String
critic_score	Average critic rating (0–10 scale)	Float
release_date	Date the game was released	Date String
total_sales	Total global sales in millions	Float
na_sales	North America sales (millions)	Float
jp_sales	Japan sales (millions)	Float
pal_sales	Europe/PAL region sales (millions)	Float
other_sales	Rest of world sales (millions)	Float

3. Data Preprocessing

3.1 Loading and Merging

Both datasets were loaded separately and merged using an inner join on the Unique_ID column. After merging, the dataset contained 64,016 records with 14 columns.

3.2 Handling Missing Values

The merged dataset had significant missing data in certain columns:

Column	Missing Count	Action Taken
critic_score	57,338	Excluded from critic-score related visualizations
last_update	46,137	Column not used in analysis
release_date	7,051	Rows excluded when analyzing yearly trends
developer	17	Filled with “Unknown”
total_sales	45,094	Rows with zero/null sales dropped

3.3 Data Cleaning and Transformation

The following operations were performed to clean and prepare the data:

- Filtered records: Rows where total_sales > 0 were retained, reducing the dataset to 17,570 records.
- String normalization: The genre column was normalized using .str.strip().str.title() to ensure consistent capitalization. The console column was converted to uppercase using .str.upper().
- Missing developer values were filled with the string "Unknown" to preserve rows for non-developer-related analyses.

3.4 Aggregation and Grouping

Several aggregated views were created to support visualization:

- genre_sales: Total sales grouped by genre, sorted descending.
- console_sales: Top 15 consoles by total sales.
- pub_sales: Top 10 publishers by total sales.
- region_totals: Sum of regional sales (NA, JP, PAL, Other).
- yearly: Year-wise total sales trend derived from release_year.

4. Visualization and Analysis

Eight visualizations were created using Matplotlib and Seaborn to explore different dimensions of the data.

4.1 Total Sales by Game Genre (Bar Chart)

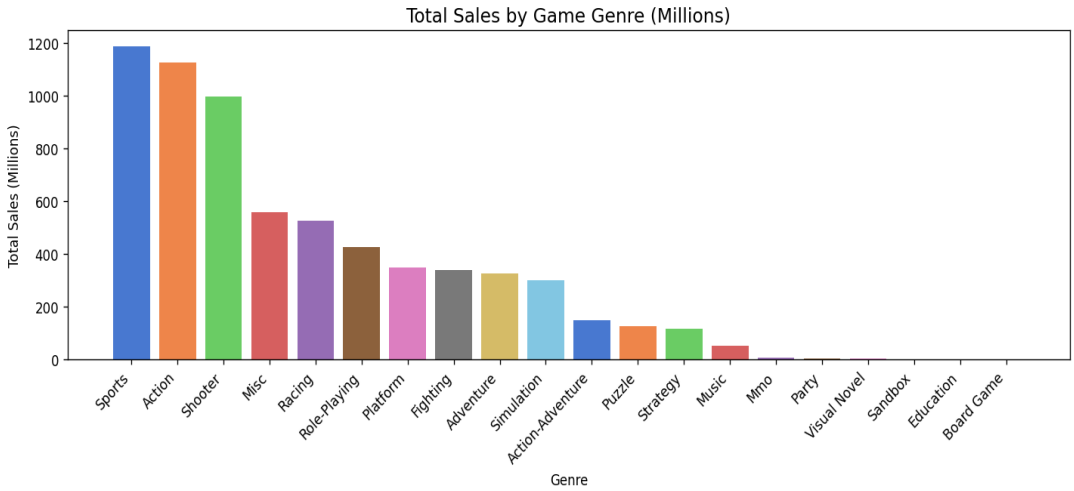

Figure 1: Total Sales by Game Genre (Millions)

Sports leads all genres in total global sales with approximately 1,187 million units, followed by Action and Shooter genres. Role-Playing and Racing games form a mid-tier category, while genres such as Puzzle, Strategy, and Adventure generate comparatively lower sales. This suggests that fast-paced and competitive genres attract the widest commercial audience.

4.2 Top 15 Consoles by Total Sales (Horizontal Bar Chart)

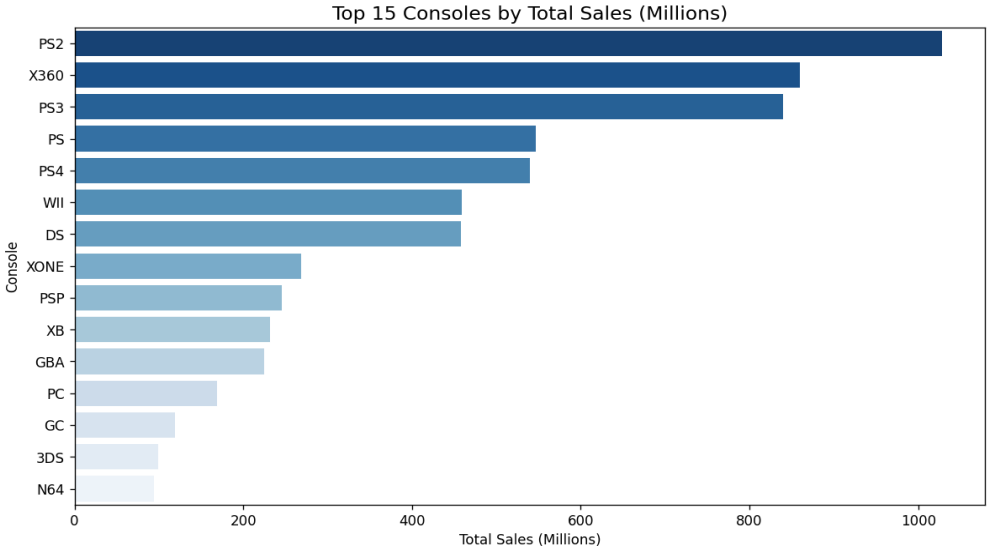

Figure 2: Top 15 Consoles by Total Sales

The PlayStation 2 (PS2) dominates as the top-selling console platform with over 1,027 million units sold across its game library. The Xbox 360, PlayStation 3, and Wii form the next tier. Sony and Microsoft platforms collectively account for the majority of sales, reflecting their strong hold on the console gaming market over the past two decades.

4.3 Regional Sales Distribution (Pie Chart)

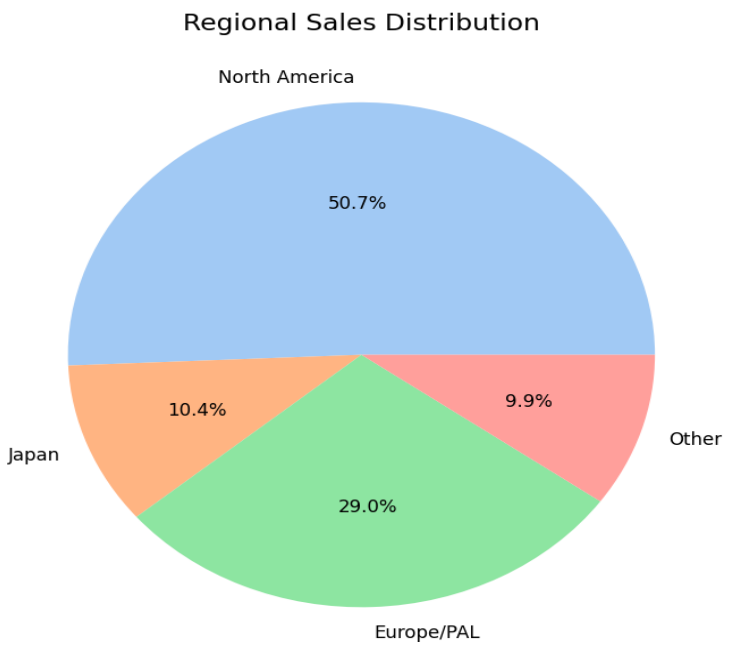

Figure 3: Regional Sales Distribution

North America is the dominant market, accounting for the largest share of global sales at approximately 51.6%. Europe/PAL follows at around 29.6%, with Japan at 10.6% and other regions contributing the remainder. This highlights the importance of the North American market for video game publishers and developers in their go-to-market strategies.

4.4 Total Sales Trend by Release Year (Line Chart)

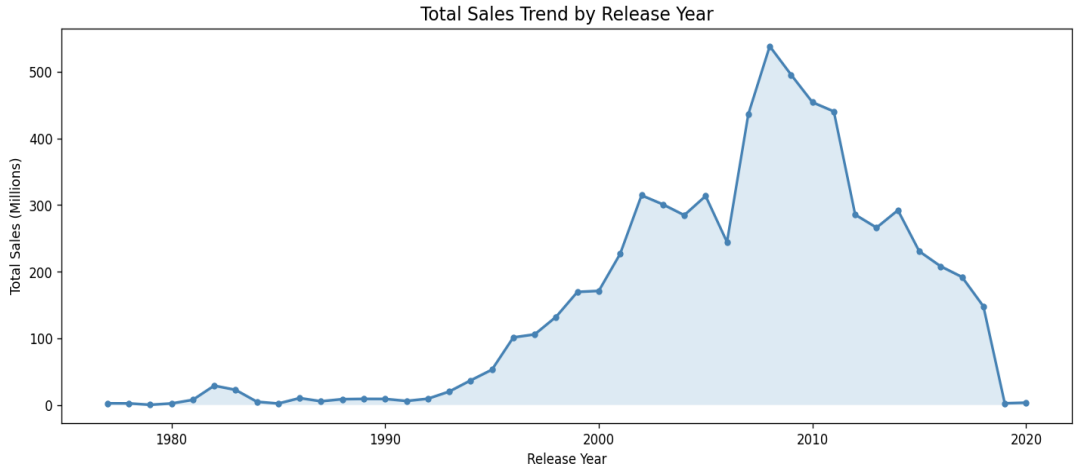

Figure 4: Total Sales Trend by Release Year

Sales were minimal throughout the 1980s and early 1990s, then accelerated significantly through the 2000s, peaking around 2008–2009. This peak corresponds with the popularity of seventh-generation consoles (PS3, Xbox 360, Wii). A sharp decline followed through the 2010s, which may be partially attributed to incomplete data for more recent titles in the dataset rather than a true market contraction.

4.5 Critic Score vs Total Sales (Scatter Plot)

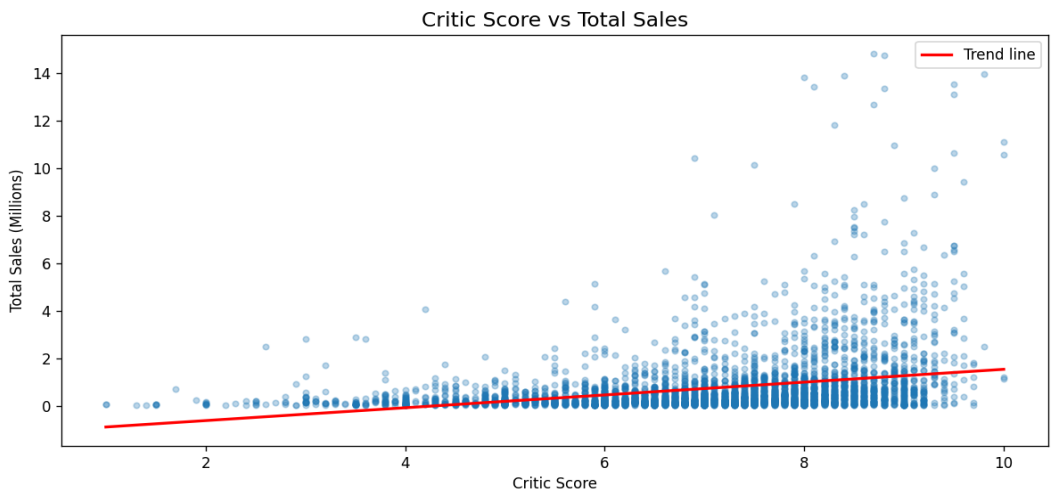

Figure 5: Critic Score vs Total Sales with Trend Line

The scatter plot reveals a weak but positive relationship between critic scores and total sales. While higher-rated games tend to sell more, there are many exceptions – indicating that factors such as brand recognition, marketing, franchise loyalty, and platform availability play a significant role beyond critical reception. The trend line confirms a slight positive slope.

4.6 Distribution of Critic Scores (Histogram)

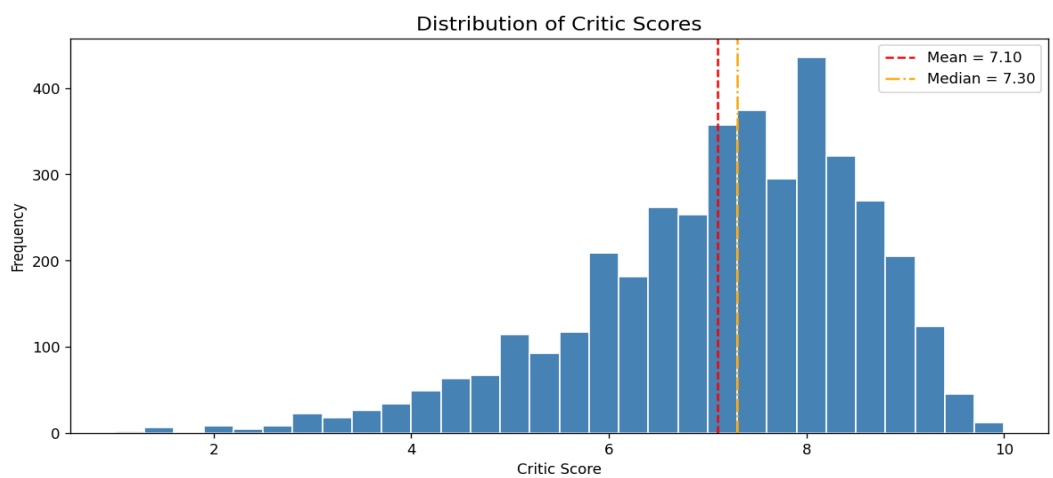

Figure 6: Distribution of Critic Scores

Critic scores follow an approximately bell-shaped (normal) distribution with a mean of 7.10 and median of 7.30. The distribution is slightly left-skewed, indicating that most reviewed games receive scores in the 6–8.5 range. Very few games score below 4 or achieve a perfect 10, suggesting that critics tend to rate games within a relatively narrow mid-to-high range.

4.7 Sales Distribution by Top 5 Genres (Box Plot)

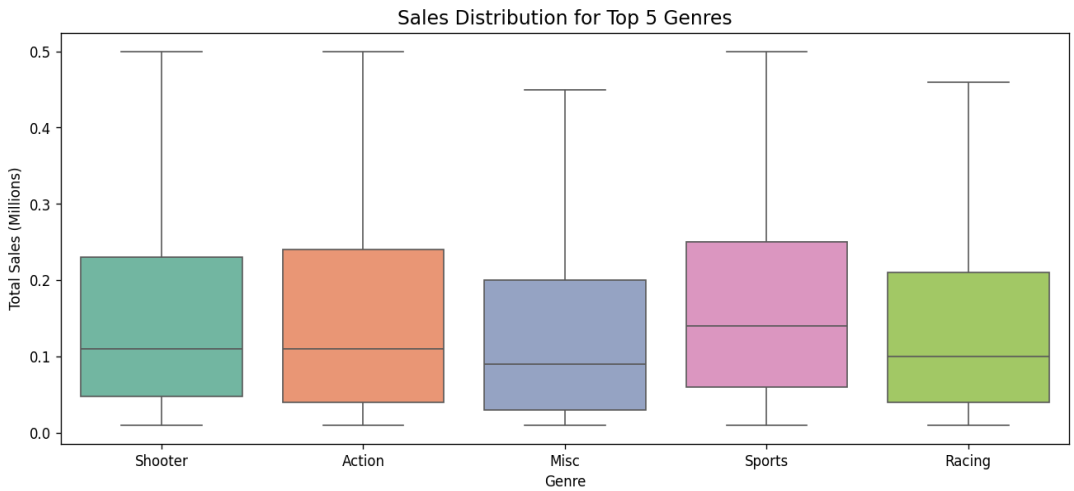

Figure 7: Sales Distribution for Top 5 Genres

The box plot reveals that most games in all top-5 genres have sales concentrated at low values (below 0.2 million units), indicating a long-tail distribution. Sports and Shooter games show slightly higher median sales values and more variability compared to Action, Role-Playing, and Racing, suggesting these genres occasionally produce blockbuster titles that significantly outperform the average.

4.8 Top 10 Publishers by Total Sales (Bar Chart)

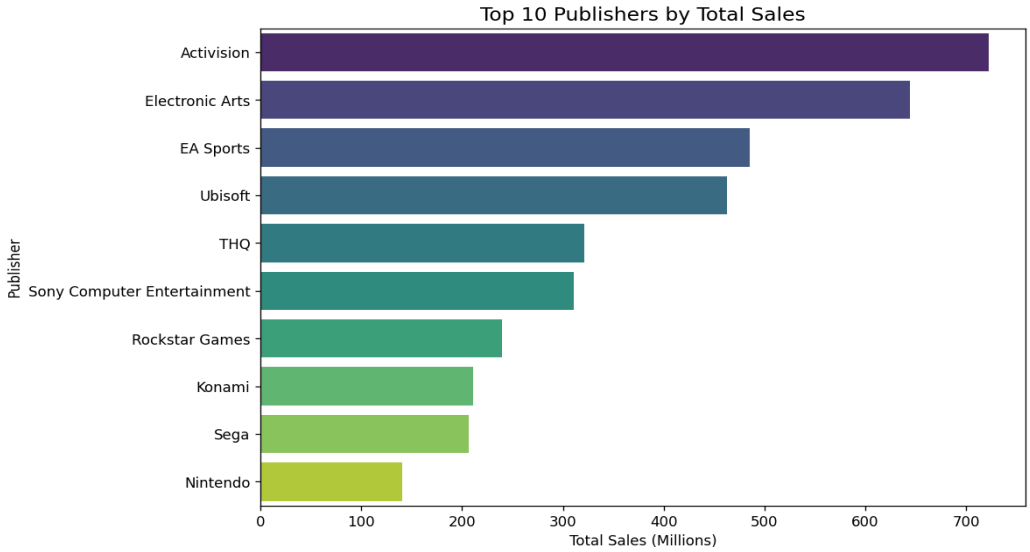

Figure 8: Top 10 Publishers by Total Sales

Activision leads among all publishers with approximately 722 million units in total sales, followed by Electronic Arts (EA) and Nintendo. These top publishers have established dominant franchises (Call of Duty, FIFA/Madden, Mario/Zelda) that consistently drive large-volume sales. The top 10 publishers collectively represent a significant portion of the total market, demonstrating concentration in the publishing industry.

5. Key Insights

The following key findings were observed from the analysis:

- **Genre Dominance**: Sports is the highest-selling genre globally, generating over 1,187 million units in total sales. Action and Shooter genres follow closely.
- **Console Dominance**: The PS2 leads all platforms with over 1,027 million units. Sony platforms (PS2, PS3, PS4, PSP) collectively dominate when combined.
- **Regional Concentration**: North America accounts for approximately 51.6% of all global game sales, making it the most critical market for publishers.
- **Sales Peak Period**: Video game sales peaked around 2008–2009, corresponding with the era of PS3, Xbox 360, and Wii. Post-2010 declines may reflect dataset gaps rather than true market decline.
- **Critic Score vs Sales**: There is a weak positive correlation between critic scores and sales. High scores do not guarantee high sales, as other factors (brand, marketing, franchise) have stronger influence.
- **Score Distribution**: Critic scores are concentrated between 6 and 8.5, with a mean of 7.10, indicating that the gaming press tends to rate games positively.
- **Long-Tail Distribution**: Sales data for all genres follows a long-tail pattern – a few blockbuster titles generate most of the revenue while the majority of games sell modestly.
- **Publisher Concentration**: Activision, EA, and Nintendo dominate publisher-level sales, highlighting the market power of major publishing houses in the industry.

6. Tools Used

The following tools and libraries were used in this project:

- **Python** – Core language used in this project
- **NumPy** – Used for numerical computations
- **Pandas** – Used for data preprocessing
- **Matplotlib and Seaborn** – Used for data visualisation by creating graphs
- **Jupyter Notebook** – Provided the coding environment for the project

7. Conclusion

This mini project provided a thorough analysis of the global video game sales landscape through the combination of two related datasets. By applying data cleaning techniques, feature engineering, grouping, and aggregation, a meaningful and well-structured dataset was prepared for visualization.

The eight visualizations created using Matplotlib and Seaborn revealed several actionable insights: Sports and Action games dominate the market; the PS2 remains the most successful gaming platform of all time by game sales volume; North America is the largest gaming market; and the mid-to-late 2000s represented the golden era of video game sales.

The weak correlation between critic scores and sales underscores that commercial success in gaming is driven by multiple interacting factors beyond critical acclaim. The long-tail distribution in genre-wise sales indicates that only a few titles achieve massive commercial success.

Overall, this analysis demonstrates the power of Python-based data analysis and visualization tools in extracting meaningful business intelligence from large, raw datasets.